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DIAGNOSTIC DEVICE, COMPUTER PROGRAM, AND DIAGNOSTIC SYSTEM

Abstract

A diagnostic device includes a reception unit and a determination unit. The reception unit is configured to receive context information and sensing information. The context information corresponds to a certain operation of a target item that constitutes a target device. The context information is a piece of a plurality of pieces of context information each describing an operation of the target item determined depending on a type of operation of the target device. The sensing information is on a physical quantity that varies in accordance with the operation of the target item. The determination unit is configured to determine a state of the target item based on the sensing information detected while the target item is performing the certain operation, and based on a model corresponding to the received context information. The model is a model of one or more models respectively defined for one or more pieces of the context information.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This is a continuation of, and claims priority under 35 U.S.C. § 120 to, U.S. application Ser. No. 17/886,786, filed on Aug. 12, 2022, which is a continuation of, and claims priority under 35 U.S.C. § 120 to, U.S. application Ser. No. 16/061,718, filed on Jun. 13, 2018, which is a national phase under 35 U.S.C. § 371 of PCT International Application No. PCT/JP2016/088471 which has an International filing date of Dec. 22, 2016, which claims priority to Japanese Application Nos. 2015-254041, filed on Dec. 25, 2015, and 2016-217144, filed on Nov. 7, 2016, the entire contents of each of which are hereby incorporated by reference.

TECHNICAL FIELD

[0002] The present invention relates to a diagnostic device, a computer program, and a diagnostic system.

BACKGROUND ART

[0003] There is known techniques for detecting abnormalities in image forming apparatuses by collecting operation sound data by a sound collecting unit and comparing the operation sound data with operation sound data prepared in advance for each operating state based on context information.

[0004] For example, PTL 1 discloses an image forming apparatus having a function of detecting an abnormal sound and specifying which unit generates the abnormal sound. The image forming apparatus compares operating sound data of each unit (e.g., drum motor, paper-feeding motor, fixing motor, and developing clutch) that has been collected and stored in advance with operation sound data collected during the operation of the image forming apparatus. When the difference between these sound data is larger than a certain level, the image forming apparatus determines the operation sound to be an abnormal sound. The image forming apparatus also specifies which unit is generating the abnormal sound based on an operation sequence table prepared for each unit.

SUMMARY OF INVENTION

Technical Problem

[0005] The technique disclosed in PTL 1 does not always reach a correct diagnosis in detecting abnormalities. In PTL 1, for example, the generation source of an abnormal sound is estimated based on an operation timing chart (see FIG. 20 in PTL 1, for example). This operation timing chart

includes operation information on each unit, but such operation information is insufficient for estimating the exact generation source. Thus, the image forming apparatus cannot always correctly estimate the sound generation source.

[0006] The present invention has been made in view of the foregoing, and it is an object of the present invention to provide a diagnostic device, a computer program, and a diagnostic system that can increase accuracy in diagnosing the operating state such as an abnormality.

Solution to Problem

[0007] According to one aspect of the present invention, a diagnostic device includes a reception unit and a determination unit. The reception unit is configured to receive context information and sensing information. The context information corresponds to a certain operation of a target item that constitutes a target device. The context information is a piece of a plurality of pieces of context information each describing an operation of the target item determined depending on a type of operation of the target device. The sensing information is on a physical quantity that varies in accordance with the operation of the target item. The determination unit is configured to determine a state of the target item based on the sensing information detected while the target item is performing the certain operation, and based on a model corresponding to the received context information. The model is a model of one or more models respectively defined for one or more pieces of the context information.

Advantageous Effects of Invention

[0008] According to the present invention, accuracy in diagnosing an abnormality can be increased.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0009] FIG. 1 is a block diagram illustrating an example of a configuration of a diagnostic system according to a first embodiment.

[0010] FIG. 2 is a block diagram illustrating an example of a hardware configuration of a process machine according to the first embodiment.

[0011] FIG. 3 is a block diagram illustrating an example of a hardware configuration of a diagnostic device according to the first embodiment.

[0012] FIG. 4 is a block diagram illustrating an example of a functional configuration of the diagnostic device according to the first embodiment.

[0013] FIG. 5 is a diagram illustrating an example of context information in association with models.

[0014] FIG. 6 is a flowchart illustrating an example of diagnostic process according to the first embodiment.

[0015] FIG. 7 is a flowchart illustrating an example of model generation process according to the first embodiment.

[0016] FIG. 8 is a diagram illustrating a specific example of generation process and diagnostic process according to the first embodiment.

[0017] FIG. 9 is a diagram illustrating an example of determination performed on a part of the context information.

[0018] FIG. 10 is a diagram illustrating an example of using the same model in another machining process.

[0019] FIG. 11 is a diagram illustrating an example of using cumulative operating time as the context information.

[0020] FIG. 12 is a diagram illustrating an example of generation of a model.

[0021] FIG. 13 is a block diagram illustrating an example of a functional configuration of a diagnostic device according to a second embodiment.

[0022] FIG. **14** is a diagram illustrating a specific example of processes according to the second embodiment.

[0023] FIG. **15** is a diagram illustrating an example of a data structure of corresponding information used for determining sensing information.

[0024] FIG. **16** is a diagram illustrating an example of a relation between the context information and a machining period.

[0025] FIG. **17** is a diagram illustrating an example of a method for specifying the machining period.

[0026] FIG. **18** is a diagram illustrating another example of the method for specifying the machining period.

[0027] FIG. **19** is a graph illustrating an example of a relation between likelihood and a determination value $r(k)$.

[0028] FIG. **20** is a diagram illustrating an example of a change in likelihood over time.

[0029] FIG. **21** is a diagram illustrating an example of using a plurality of thresholds.

[0030] FIG. **22** is a diagram illustrating an example of using a plurality of thresholds.

[0031] FIG. **23-1** is a diagram illustrating an example of a determination method according to a sixth modification.

[0032] FIG. **23-2** is a diagram illustrating another example of the determination method according to the sixth modification.

[0033] FIG. **23-3** is a diagram illustrating still another example of the determination method according to the sixth modification.

[0034] FIG. **24-1** is a diagram illustrating an example of a determination method according to a seventh modification.

[0035] FIG. **24-2** is a diagram illustrating another example of the determination method according to the seventh modification.

[0036] FIG. **24-3** is a diagram illustrating still another example of the determination method according to the seventh modification.

DESCRIPTION OF EMBODIMENTS

[0037] Exemplary embodiments of a diagnostic device, a computer program, and a diagnostic system according to the present invention are described in detail below with reference to the accompanying drawings.

First Embodiment

[0038] FIG. **1** is a block diagram illustrating an example of a configuration of a diagnostic system according to a first embodiment. As illustrating in FIG. **1**, this diagnostic system includes a process machine **200** and a diagnostic device **100**. The process machine **200** is an example of a target device of diagnosis performed by the diagnostic device **100**.

[0039] The process machine **200** and the diagnostic device **100** may be connected with each other in any form. For example, the process machine **200** and the diagnostic device **100** may be connected with each other via a dedicated connection line, a wired network such as a wired local area network (LAN), or a wireless network.

[0040] The process machine **200** includes a numerical control unit **201**, a communication control unit **202**, and a machine tool **203**. The machine tool **203** includes a sensor **211**, a drive unit **212**, and a tool **213**.

[0041] The machine tool **203** is a machine that processes a processing target under the control of the numerical control unit **201**. The machine tool **203** includes the drive unit **212** that operates under the control of the numerical control unit **201**. The drive unit **212** is, for example, a motor. The tool **213** is a target item that is actually driven by the drive unit **212**. The tool **213** may be any type of tool that can be used for machining and can be controlled by numerical control, such as a drill or an end mill that processes a target workpiece. The machine tool **203** may include one or more drive units **212**.

[0042] The numerical control unit **201** controls the machine tool **203** to process a workpiece by numerical control. The numerical control unit **201**, for example, generates and outputs numerical control data for controlling the operation of the drive unit **212**. The numerical control unit **201** outputs context information to the communication control unit **202**. The context information is information defined for each type of operation of the process machine **200**. The context information defined for each type of operation includes a plurality of pieces of context information. The context information includes, for example, information for identifying the tool **213** driven by the drive unit **212**, the number of rotations of the drive unit **212**, the rotating speed of the drive unit **212**, and moving information on the drive unit **212** and the tool **213**.

[0043] The numerical control unit **201** transmits the context information indicating, for example, the current operation to the diagnostic device **100** via the communication control unit **202**. Depending on the procedure of a machining process in which a processing target is processed, the numerical control unit **201** changes the types of the tool **213** driven by the drive unit **2** and the drive state of the drive unit **212** (e.g., the number of rotations and rotation speed). Every time the numerical control unit **201** changes the types of operation, the numerical control unit **201** transmits the context information corresponding to the type of the operation after the change to the diagnostic device **100** via the communication control unit **202**.

[0044] The communication control unit **202** controls the communication between the process machine **200** and an external device such as the diagnostic device **100**. For example, the communication control unit **202** transmits the context information corresponding to the current operation to the diagnostic device **100**.

[0045] The sensor **211** is a sensing unit that detects a physical quantity that varies in accordance with the operation of the process machine **200** and outputs sensing information (sensor data). The sensor **211** may be any type of sensors, and may detect any type of physical quantity. For example, the sensor **211** may be a microphone that outputs acoustic data, an acceleration sensor that outputs acceleration data, or an acoustic emission (AE) sensor that outputs AE wave data, as the sensing information. Any number of sensors **211** may be included in the machine tool **203**. The machine tool **203** may include a plurality of sensors **211** that detect the same physical quantity, or may include a plurality of sensors **211** that detect different types of physical quantities.

[0046] If, for example, an edge of the tool **213** breaks or chips during use, the sound that the tool **213** is making in the machining process changes. In this case, the sensor **211** (microphone) detects acoustic data and the detected data is compared with, for example, a model representing a normal sound. Thus, such an abnormality in the operation of the process machine **200** can be detected.

[0047] The diagnostic device **100** includes a communication control unit **101** and a determination unit **102**. The communication control unit **101** controls the communication between the diagnostic device **100** and an external device such as the process machine **200**. The communication control unit **101**, for example, receives the context information and the sensing information from the process machine **200**. The determination unit **102** refers to the context information and the sensing information to determine whether the operation of the process machine **200** is normal. Details of each unit will be described later.

[0048] FIG. 2 is a block diagram illustrating an example of a hardware configuration of the process machine **200**. As illustrated in FIG. 2, the process machine **200** includes a central processing unit (CPU) **51**, a read only memory (ROM) **52**, a random access memory (RAM) **53**, a communication I/F **54**, a drive control circuit **55**, and a motor **56** that are connected with each other via a bus **58**.

[0049] The CPU **51** controls the entire operation of the process machine **200**. The CPU **51** loads a computer program stored in the ROM **52** or other storage units onto, for example, the RAM **53** as a work area and executes the computer program to control the entire operation of the process machine **200**, thereby implementing its machining function.

[0050] The communication I/F **54** is an interface for communicating with an external device such as the diagnostic device **100**. The drive control circuit **55** is a circuit for controlling the motor **56** to

drive. The motor **56** drives the tool **213** such as a drill, a cutter, and a table that are used in a machining process. The motor **56** corresponds to, for example, the drive unit **212** in FIG. **1**. A sensor **57** is installed in the process machine **200**. The sensor **57** detects a physical quantity that varies depending on the operation of the process machine **200** and outputs the sensing information to the diagnostic device **100**. The sensor **57** corresponds to, for example, the sensor **211** in FIG. **1**. [0051] The numerical control unit **201** and the communication control unit **202** in FIG. **1** may be implemented by causing the CPU **51** in FIG. **2** to execute a computer program, that is, implemented by software, or may be implemented by hardware such as an integrated circuit (IC), or may be implemented by both software and hardware.

[0052] FIG. **3** is a block diagram illustrating an example of a hardware configuration of the diagnostic device **100**. As illustrated in FIG. **3**, the diagnostic device **100** includes a CPU **61**, a ROM **62**, a RAM **63**, a communication I/F **64**, and a hard disk drive (HDD) **65** that are connected with each other via a bus **66**.

[0053] The CPU **61** controls the entire operation of the diagnostic device **100**. The CPU **61** loads a computer program stored in the ROM **62** or other storage units onto, for example, the RAM **63** as a work area and executes the computer program to control the entire operation of the diagnostic device **100**, thereby implementing its diagnostic function. The communication I/F **64** is an interface for communicating with an external device such as the process machine **200**. The HDD **65** stores therein information such as setting information of the diagnostic device **100**, the context information transmitted from the numerical control unit **201** via the communication control unit **202** of the process machine **200**, and the sensing information transmitted from the sensor **211**. The diagnostic device **100** may include a non-volatile storage unit such as an electrically erasable programmable ROM (EEPROM) or a solid state drive (SSD) instead of or in addition to the HDD **65**.

[0054] FIG. **4** is a block diagram illustrating an example of a functional configuration of the diagnostic device **100**. As illustrated in FIG. **4**, the diagnostic device **100** includes an acceptance unit **103**, a feature extraction unit **104**, a generation unit **105**, and a storage unit **111** in addition to the communication control unit **101** and the determination unit **102** described above.

[0055] The storage unit **111** stores therein various types of information that are needed for the diagnostic function of the diagnostic device **100**. The storage unit **111** can be implemented by, for example, the RAM **63** and the HDD **65** in FIG. **3**. The storage unit **111** stores therein, for example, one or more models to be used for the determination of abnormalities.

[0056] Such models are generated by learning based on, for example, the sensing information that has been detected during a normal operation of the process machine **200**. Alternatively, depending on a purpose of detection, a model is generated such that a new tool **213** is attached to the process machine **200** and is used in a certain machining process until the tool **213** is in an abnormal state (until the edge of the tool **213** breaks or chips, for example) and data for the model is acquired. Learning method may be any method, and model format may be any format. For example, models such as a Gaussian mixture model (GMM) and a hidden Markov model (HMM) can be used, and their corresponding model learning methods can be applied.

[0057] Alternatively, for example, a certain machining term after a new tool **213** is attached may be used as a learning term, in which a normal state and an abnormal state of the tool **213** may be regularized, and the regularized criteria may be stored in the storage unit **111** as models. For example, the learning term may be determined in the following manner. A new tool **213** is attached and the first ten times of machining processes are determined to be a learning term for regularizing the diagnostic criteria. Diagnostic criteria may be regularized in another device before the actual machining process, and the predetermined criteria may be stored in the storage unit **111** as models.

[0058] In the first embodiment, a model is generated for each type of context information. The storage unit **111** stores therein, for example, context information and a model corresponding to the context information in association with each other. FIG. **5** is a diagram illustrating an example of

context information in association with models. As illustrated in FIG. 5, a model is generated for, for example, each type of the tool **213** to be attached to the process machine **200**. When the tool **213** is a drill, a model is generated for each diameter of the drill. When three diameters of 1 mm, 3 mm, and 5 mm are available, models are generated for the respective diameters. When the determination unit **102** recognizes the diameter of the drill to be used for a machining process based on the context data received from the process machine **200**, the determination unit **102** selects a learned model corresponding to the diameter of the drill and uses the model for determination.

[0059] The communication control unit **101** includes a reception unit **101a** and a transmission unit **101b**. The reception unit **101a** receives various types of information transmitted from an external device such as the process machine **200**. For example, the reception unit **101a** receives context information corresponding to the current operation of the process machine **200** and sensing information transmitted from the sensor **211**. The transmission unit **101b** transmits various types of information to the external device.

[0060] The feature extraction unit **104** extracts, from the sensing information, feature information (feature) to be used for model generation process and for the determination performed by the determination unit **102**. The feature information may be any type of information that represents a feature of the sensing information. When, for example, the sensing information is acoustic data collected by a microphone, the feature extraction unit **104** may extract features such as energy, frequency spectrum, and mel-frequency cepstrum coefficients (MFCC).

[0061] The generation unit **105** generates models for determining a normal operation by learning based on the feature information extracted from the sensing information detected during a normal operation. When the models are generated in an external device, the diagnostic device **100** does not need to include the generation unit **105**. When context information for which no model has been defined and sensing information corresponding to the context information are input, the generation unit **105** may use feature information extracted from this sensing information and generate a model corresponding to this context information.

[0062] Although models are generated by the generation unit **105**, the models may be generated in an external device and the generated models may be received via the communication I/F **64** and stored in the HDD **65** for use.

[0063] The determination unit **102** determines the state of the operation of the process machine **200** based on the sensing information received by the reception unit **101a** and the model corresponding to the received context information. The state of the operation of the process machine **200** indicates, for example, whether the process machine **200** is operating in a normal state. For example, the determination unit **102** requests the feature extraction unit **104** to extract feature information from the sensing information. The determination unit **102** calculates a likelihood representing how likely the feature information extracted from the sensing information indicates normality based on a corresponding model. The determination unit **102** compares the likelihood with a threshold that has been determined in advance. When, for example, the likelihood is equal to or larger than the threshold, the determination unit **102** determines that the process machine **200** is operating in a normal state. When the likelihood is less than the threshold, the determination unit **102** determines that the process machine **200** is operating in an abnormal state. The feature varies in accordance with the state of the tool **213**. Thus, the determination performed by the determination unit **102** includes determination of an index indicating, for example, how much the tool **213** is worn to determine the state of the operation of the process machine **200**.

[0064] The method for determining whether the operation is in a normal state is not limited to this. The determination method may be any method that can determine whether the operation is in a normal state based on sensing information and a model. For example, the determination unit **102** may compare a value indicating a change in the likelihood with a threshold instead of directly comparing a value of the likelihood with a threshold.

[0065] The acceptance unit **103** receives an input of context information that differs from the context information received by the reception unit **101a** from the process machine **200**. For example, the acceptance unit **103** may be configured to acquire cumulative operating time from the process machine **200**. In this case, the process machine **200** may include, for example, a function of resetting (initialization) the cumulative operating time that functions when the process machine **200** changes tools **213**.

[0066] Alternatively, the acceptance unit **103** may receive the cumulative operating time not from the process machine **200**. The acceptance unit **103** receives context information input from an operating unit such as a keyboard or a touch panel. The context information received by the acceptance unit **103** is not limited to the cumulative operating time. The context information may be, for example, information on specifications of the tool **213** to be used (diameter of a cutting edge, the number of edges, material of the tool **213**, and presence or absence of coating on the tool **213**), or information on the workpiece to be processed (material, for example). The acceptance unit **103** may receive the context information from an external device such as a server device or a personal computer. If the diagnostic device **100** has no need for receiving the context information from devices other than the process machine **200**, the diagnostic device **100** does not need to include the acceptance unit **103**.

[0067] The units in FIG. **4** (the communication control unit **101**, the determination unit **102**, the acceptance unit **103**, the feature extraction unit **104**, and the generation unit **105**) may be implemented by causing the CPU **61** in FIG. **3** to execute a computer program, that is, implemented by software, or may be implemented by hardware such as an IC, or may be implemented by both software and hardware.

[0068] Described next is the diagnostic process performed by the diagnostic device **100** according to the first embodiment with reference to FIG. **6**. FIG. **6** is a flowchart illustrating an example of the diagnostic process according to the first embodiment.

[0069] As described above, the numerical control unit **201** of the process machine **200** sequentially transmits the context information indicating the current operation to the diagnostic device **100**. The reception unit **101a** receives the context information transmitted from the process machine **200** (Step **S101**). The sensor **211** of the process machine **200** sequentially outputs sensing information during a machining process. The reception unit **101a** receives the sensing information (sensor data) transmitted from the process machine **200** (Step **S102**).

[0070] The feature extraction unit **104** extracts feature information from the received sensing information (Step **S103**). The determination unit **102** determines whether the process machine **200** is operating in a normal state based on the extracted feature information and a model corresponding to the received context information (Step **S104**). The determination unit **102** outputs a result of the determination (Step **S105**). The result of the determination may be output by any method. When, for example, the diagnostic device **100** includes a display device such as a display, the determination unit **102** may display the result of the determination on this display device. The determination unit **102** may output the result of the determination to an external device such as a server device or a personal computer.

[0071] Described next is model generation process performed by the diagnostic device **100** according to the first embodiment with reference to FIG. **7**. FIG. **7** is a flowchart illustrating an example of the model generation process according to the first embodiment. The model generation process is performed, for example, before the diagnostic process. As described above, the diagnostic device **100** may be configured to perform the model generation process upon receiving context information for which no model has been defined. When models are generated in an external device as described above, the diagnostic device **100** does not need to perform the model generation process.

[0072] The reception unit **101a** receives context information transmitted from the process machine **200** (Step **S201**). The reception unit **101a** receives sensing information (sensor data) transmitted

from the process machine **200** (Step S202).

[0073] The received context information and sensing information are used for generating a model. A model is generated for each type of context information, and thus, sensing information needs to be associated with the corresponding context information. The reception unit **101a**, for example, associates the received sensing information with context information received at the same timing and stores them in, for example, the storage unit **111**. The diagnostic device **100** may temporarily store context information and sensing information in, for example, the storage unit **111**, and determine whether the information comes from a normal operation. The diagnostic device **100** may generate a model based only on the information determined to be information on a normal operation. In other words, the diagnostic device **100** may generate a model based on sensing information that has been labeled as normal.

[0074] The determination (labeling) as to whether the process machine **200** is operating in a normal state can be performed at any timing after the information is stored in, for example, the storage unit **111**, or in real time with the process machine **200** kept operating. Instead of performing the labeling, the diagnostic device **100** may generate a model on the assumption that the information indicates that the process machine **200** is operating in a normal state. If the information that has been assumed to indicate a normal state indicates actually an abnormal state, the diagnostic device **100** fails to perform a correct determination based on the generated model. This situation may cause the diagnostic device **100** to determine abnormalities more often, which, however, allows the diagnostic device **100** to recognize this incorrectly generated model and delete it, for example. The diagnostic device **100** may use a model generated from information indicating an abnormal state as a model for determining an abnormality of the process machine **200**.

[0075] The feature extraction unit **104** extracts feature information from the collected sensing information (Step S203). The generation unit **105** generates a model corresponding to the context information based on the feature information extracted from the sensing information associated with this context information (Step S204). The generation unit **105** stores the generated model in, for example, the storage unit **111** (Step S205).

[0076] The following describes specific examples of the model generation process and the diagnostic process according to the first embodiment. FIG. **8** is a diagram illustrating a specific example of the processes according to the first embodiment.

[0077] FIG. **8** illustrates, for example, the model generation process and the diagnostic process performed on a part of a machining process for processing a certain workpiece. In the model generation process, the diagnostic device **100** uses a plurality of pieces of sensing information (sensing information **711a** to **711c** in FIG. **8**) received together with context information **701**. The number of the pieces of the sensing information is not limited to three, but may be any number.

[0078] The context information **701** indicates that this machining process includes operations to drive four motors (motor A, motor B, motor C, and motor D). The feature extraction unit **104** extracts feature information from the received sensing information. The generation unit **105** generates a model for each piece of context information corresponding to the motor based on the feature information extracted from the corresponding sensing information. The generated models are stored in, for example, the storage unit **111**. FIG. **8** illustrates an example of storing, in the storage unit **111**, a model (motor B) generated corresponding to the context information indicating that the motor B is driving. The stored model will be referred to in the diagnostic process to be performed later.

[0079] In the diagnostic process, the diagnostic device **100** receives the context information **701** and sensing information **721** in the same manner as in the model generation process. When the context information indicates that the motor B is driving, the determination unit **102** determines whether the process machine **200** is operating in a normal state based on the sensing information received during a term in which this context information is received and based on the model “motor B” stored in the storage unit **111**.

[0080] When other context information is received, the determination unit **102** performs the determination based on the sensing information and the model corresponding to this context information in the same manner as described above. The determination unit **102** does not necessarily perform the determination on all the context information. FIG. **9** is a diagram illustrating an example of determination performed on a part of the context information.

[0081] In the example of FIG. **9**, a model is generated only when the context information indicates that the motor B is driving. The diagnostic process is performed when context information **701-2** indicating that the motor B is driving is received. This configuration enables the diagnostic device **100** to perform the diagnostic process based only on the sensing information that is effective in determining an abnormality. When, for example, acoustic data is used as sensing information, the machining process may include a period, such as a silent period, on which no determination is needed. Eliminating such unnecessary period from the periods subjected to the determination can reduce incorrect determination and can reduce the calculation costs. In other words, this configuration achieves more accurate and more efficient diagnostic process.

[0082] When, for example, the process machine **200** uses the same motor in a different machining process, the diagnostic device **100** may perform the diagnostic process based on the same model corresponding to the motor. FIG. **10** is a diagram illustrating an example of using the same model in another machining process.

[0083] Context information **901** in FIG. **10** indicates that this machining process includes operations to drive four motors (motor X, motor Y, motor Z, and motor B). This machining process uses the motor B, which is common to the machining process illustrated in, for example, FIG. **8**. Thus, the determination unit **102** can perform the diagnostic process based on the same model “motor B” and sensing information **921** in the machining process in FIG. **10**.

[0084] FIG. **11** is a diagram illustrating an example of using cumulative operating time as context information. Context information **1001** indicates cumulative operating time of each motor. In the example of FIG. **11**, a model is generated for each type of the motors in each range (e.g., 0 to 6 months and 6 to 12 months) of the cumulative operating time. For example, a storage unit **111a** stores therein models for the motors having cumulative operating time of 0 to 6 months. A storage unit **111b** stores therein models for the motors having cumulative operating time of 6 to 12 months. A storage unit **111c** stores therein models for the motors having cumulative operating time of 12 months and more. The storage units **111a** to **111c** may be implemented by physically different storage media, or a physically identical storage medium. The determination unit **102** uses the context information **701** and the context information **1001** so as to specify a corresponding model, and performs the diagnostic process based on the specified model.

[0085] FIG. **12** is a diagram illustrating an example of generating a model when context information for which no model has been defined is input. Context information **1101** in FIG. **12** indicates that this machining process includes operations to drive four motors (motor X, motor C, motor D, and motor B). Suppose that no model has been generated for context information indicating that the motor X is driving.

[0086] In this case, the feature extraction unit **104** extracts feature information from sensing information detected in a term corresponding to the context information indicating that the motor X is driving. The feature information is extracted from a plurality of pieces of sensing information (in FIG. **12**, sensing information **1111a**, **1111b**, and **1111c**). The generation unit **105** generates a model “motor X” corresponding to the context information indicating that the motor X is driving based on the extracted feature information and stores the generated model in the storage unit **111**. Thus, the diagnostic device **100** can determine an abnormality in a future term in which the motor X is driving.

[0087] According to the first embodiment, the diagnostic device **100** receives, from the process machine **200**, context information indicating the current operation of the process machine **200**, and uses a model corresponding to the received context information, and the diagnostic device **100** can

determine an abnormality. This configuration enables the diagnostic device **100** to highly accurately specify the drive unit currently driving and highly accurately diagnose an abnormality.

Second Embodiment

[0088] In the first embodiment, the diagnostic device **100** determines whether the process machine **200** is operating in a normal state based on a single type of sensing information. The number of types of the sensing information used in the determination is not limited to one, and may be two or more. A diagnostic system according to a second embodiment determines an abnormality in the process machine **200** based on a plurality of types of sensing information.

[0089] The diagnostic system according to the second embodiment has the same configuration as that of the first embodiment illustrated in FIG. **1**, and thus the explanation thereof is omitted. FIG. **13** is a block diagram illustrating an example of a functional configuration of a diagnostic device **100-2** according to the second embodiment. As illustrated in FIG. **13**, the diagnostic device **100-2** includes the communication control unit **101**, a determination unit **102-2**, the acceptance unit **103**, the feature extraction unit **104**, the generation unit **105**, and the storage unit **111**.

[0090] The second embodiment differs from the first embodiment in that the determination unit **102-2** has a different function. Other units of the diagnostic device **100-2** have the same configurations and functions as those of the diagnostic device **100** according to the first embodiment as illustrated in the block diagram in FIG. **4**, and thus are labeled with the same reference signs, and the explanation thereof is omitted.

[0091] The determination unit **102-2** determines whether the process machine **200** is operating in a normal state based on a plurality of types of sensing information. For example, the determination unit **102-2** changes the types of sensing information to be used for the determination depending on the context information.

[0092] FIG. **14** is a diagram illustrating a specific example of processes according to the second embodiment. In the example of FIG. **14**, the diagnostic device **100-2** receives context information **1201** and a plurality of types of sensing information **1221a** and **1221b**. The sensing information **1221a** is, for example, acoustic data. The sensing information **1221b** is, for example, acceleration data.

[0093] When the context information indicates that the motor A is driving, the determination unit **102-2** performs determination based on the feature information extracted from sensing information **1222b**, of the sensing information **1221b**, detected in a term corresponding to this context information. When the context information indicates that the motor B is driving, the determination unit **102-2** performs determination based on the feature information extracted from sensing information **1222a**, of the sensing information **1221a**, detected in a term corresponding to this context information. The determination unit **102-2** reads a model corresponding to this context information from a storage unit **111a-2** or a storage unit **111b-2** and uses the model.

[0094] The sensing information corresponding to the context information may be determined based on the corresponding information stored in, for example, the storage unit **111**. FIG. **15** is a diagram illustrating an example of a data structure of the corresponding information used for determining the sensing information. The corresponding information includes, for example, sensor data and context information. The determination unit **102-2** can determine the sensing information corresponding to the context information with reference to the corresponding information.

[0095] Described next are modifications that are applicable to the first and the second embodiments.

First Modification

[0096] The context information only indicates the period in which a certain drive unit **212** is driving. Thus, the diagnostic device may fail to precisely extract the actual machining period in which, for example, the tool **213** driven by the drive unit **212** is rotating in contact with a workpiece and machining it. In other words, an abnormality may be determined less accurately.

[0097] FIG. **16** is a diagram illustrating an example of a relation between context information and a

machining period. Context information **1501** indicates that the motor B is driving. When the determination unit **102** only refers to the context information **1501**, it specifies sensing information corresponding to a waveform period **1512** in the sensing information. However, the period in which the tool **213** is actually machining the workpiece is a waveform period **1511**. When, for example, the sensing data is acoustic data, the waveform period **1511** corresponds to a period of acoustic data detected while the tool **213** is in contact with the workpiece and generating a sound.

[0098] In a first modification, the determination unit **102** is configured to specify an actual machining period based on both context information and sensing information. In other words, the determination unit **102** determines a term used for the determination from a term specified by the received context information based on the received sensing information. The determination unit **102** performs determination based on the sensing information in the determined term and a model.

[0099] For example, the determination unit **102** determines a machining period by specifying the time at which the feature of the sensing information changes. FIG. **17** is a diagram illustrating an example of a method for specifying a machining period. In the example of FIG. **17**, the determination unit **102** specifies a time point **1601** at which the amplitude of the sensing information changes to exceed thresholds (e.g., “a” and “-a”) that have been set in advance and specifies a time point **1602** at which the amplitude changes to fall within the thresholds. The determination unit **102** extracts a period **1611** between the time point **1601** and the time point **1602**. The determination unit **102** performs determination by using the feature information extracted from the sensing information in the period **1611**. Using the sensing information in a period corresponding to the actual machining period enables the determination unit **102** to perform determination more accurately.

Second Modification

[0100] In a second modification, the feature information is extracted (machining period is specified) by using a period specifying signal. The period specifying signal is generated in the diagnostic device **100** based on the context information transmitted from the process machine **200**, but the signal may be generated in the process machine **200** and transmitted to the diagnostic device **100**.

[0101] FIG. **18** is a diagram illustrating an example of a method for specifying a machining period in the second modification. When a period specifying signal **1801** indicates a machining period (a period in which a high-level signal is being output, for example), the determination unit **102** uses feature information extracted from sensing information detected in the corresponding period for the determination. In the example of FIG. **18**, the determination unit **102** determines periods **1811**, **1813**, and **1814** not to be the machining periods, and determines a period, in a period **1812**, corresponding to the signal **1801** that indicates a machining period (a period in which a high-level signal is being output, for example) to be the machining period.

[0102] When, for example, the tool **213** is a drill and is used for making a hole in a processing target, the diagnostic device **100** generates a period specifying signal based on the context information indicating the drill feed speed toward the processing target. When the diagnostic device **100** detects, from the context information, a cutting drill feed speed, which indicates that the drill has started drilling the processing target, the diagnostic device **100** generates a high-level period specifying signal (indicating that the tool **213** is processing the subject). When the diagnostic device **100** detects a change in the context information indicating that the cutting drill feed speed is changed to a non-cutting drill feed speed, which indicates that the drill has finished making a hole in the processing target, the diagnostic device **100** outputs a low-level period specifying signal instead of the high-level signal (indicating that the tool **213** is not processing the subject).

[0103] As another example, the diagnostic device **100** may generate the period specifying signal based on coordinates information indicating machining locations as the context information. In this method, the diagnostic device **100** may generate the period specifying signal based on the context

information indicating that the drill or a processing target has moved to a machining coordinate location and the context information indicating that the drill or the processing target has moved away from the machining coordinate location. The diagnostic device **100** may use any type of context information that can specify a machining period in which the tool **213** is actually processing a processing target.

[0104] When the period specifying signal is generated in the process machine **200**, the process machine **200** is installed with a computer program that allows the process machine **200** to generate a high-level period specifying signal under the same conditions as those of the diagnostic device **100**.

[0105] As described above, the diagnostic device **100** generates a model based only on the sensing information detected during a term in which a high-level period specifying signal is being output, and not on the whole sensing information detected throughout the term in which the motor B is driving. The diagnostic device **100** diagnoses the state of the tool **213** based only on the sensing information detected during a period in which the tool **213** is actually processing the processing target. This configuration enables the diagnostic device **100** to determine the state of the tool **213** (whether the edge of the drill is abraded or cracked, for example) based only on the sensing information detected during a period in which the tool **213** is actually in contact with the tool **213**, thereby increasing the accuracy of diagnosis.

Third Modification

[0106] The diagnostic device **100** may use a value of likelihood or a value representing a change in likelihood in determining whether the tool **213** is operating in a normal state. In a third modification, described is an example of a value representing a change in likelihood. As a value representing a change in likelihood, a variance of likelihood can be used, for example. A variance of likelihood in a period X is calculated, for example, by Equation (1) below, where $n_{\text{sub.X}}$ represents the number of frames in the period X, k represents an index of the frames, $X_{\text{sub.i}}$ represents the likelihood in a frame i (frame likelihood), and $\mu_{\text{sub.X}}$ represents the mean of the frame likelihoods in the period X. Each frame corresponds to a unit period for which the likelihood is calculated.

$$[00001] \quad V_X(k) = \frac{1}{n_X} \cdot \text{Math.} \sum_{i=k - \frac{n_X-1}{2}}^{K + \frac{n_X-1}{2}} (x_i - \mu_X)^2 \quad (1)$$

[0107] A scored value may be calculated based on the variance calculated by Equation (1), for example, and may be used for the determination. For example, $r(k)$ in Equation (2) below may be used for the determination.

$$[00002] \quad r(k) = V_S(k) / V_L(k) \quad (2)$$

[0108] S and L each represent the type of the period X. S represents a short period, and L represents a long period. $V_{\text{sub.S}}(k)$ represents a value calculated by Equation (1) for a short period. $V_{\text{sub.L}}(k)$ represents a value calculated by Equation (1) for a long period. With respect to S and L, $n_{\text{sub.S}}$ and $\mu_{\text{sub.S}}$, and $n_{\text{sub.L}}$ and $\mu_{\text{sub.L}}$ corresponding to $n_{\text{sub.X}}$ and $\mu_{\text{sub.X}}$, respectively, are calculated.

[0109] FIG. **19** is a graph illustrating an example of a relation between the likelihood and the determination value $r(k)$. In a determination method that compares a likelihood with a threshold, which is assumed to be, for example, 0.97, the diagnostic device **100** determines that the operation from 200 to 300 frames is performed in a normal state. In a determination method that compares a determination value $r(k)$ with a threshold (e.g., 1.0), the diagnostic device **100** can determine that the operation from 200 to 300 frames is performed in an abnormal state.

Fourth Modification

[0110] The determination unit **102** may be configured to determine and output a change in the likelihood over time as well as compare a likelihood with a threshold in determining whether the operation is performed in a normal state. When, for example, the determination unit **102** uses a

single model representing a normal operation, the determination unit **102** may store calculated likelihoods in, for example, the storage unit **111** and obtain a change in the likelihood (whether the likelihood is decreasing over time, for example).

[0111] FIG. **20** is a diagram illustrating an example of a change in the likelihood over time. The determination unit **102** can obtain a change in the likelihood over time by storing, for example, time of the determination and likelihoods. The determination unit **102** determines that the operation is performed in an abnormal state when the likelihood falls below the threshold, which is the same configuration as that in the embodiments above. The determination unit **102** may be configured to obtain and output, for example, information indicating a change in the likelihood over time and time at which the likelihood is anticipated to fall below the threshold.

Fifth Modification

[0112] The determination unit **102** may be configured to determine quality ranks or quality levels based on a plurality of thresholds. FIGS. **21** and **22** are diagrams each illustrating an example of using a plurality of thresholds. FIGS. **21** and **22** each illustrate an example of using four thresholds (thresholds **1** to **4**), but the number of thresholds is not limited to four.

[0113] FIG. **21** illustrates an example of five quality ranks from R**1** to R**5** that are separated based on whether the likelihood is above or below the four thresholds. The determination unit **102** compares the likelihood with the thresholds, and determines and outputs the quality rank. The quality rank can be used as, for example, information (quality information) indicating quality of an object (processed item) machined by the process machine **200**.

[0114] FIG. **22** illustrates an example of determining whether the quality is at a normal level (no problem), at a caution level, or at an abnormal level (below a quality standard) based on whether the likelihood is above or below the four thresholds. When the determination unit **102** determines that the quality is at a normal level, the determination unit **102** classifies the quality into further two levels, L**11** and L**12**. When the determination unit **102** determines that the quality is at a caution level, the determination unit **102** classifies the quality into further two levels, L**21** and L**22**. The determination unit **102** may classify the quality into three or more levels.

Sixth Modification

[0115] The determination unit **102** may calculate a plurality of likelihoods based on a plurality of models that are defined in accordance with a quality rank to determine whether the operation is performed in a normal state, and to determine the quality level when determining a normal operation. FIGS. **23-1** to **23-3** are diagrams illustrating an example of a determination method according to a sixth modification.

[0116] FIGS. **23-1**, **23-2**, and **23-3** correspond to the quality ranks R**1**, R**2**, and R**3**, respectively, and each illustrate an example of the likelihood calculated based on a model used for determining the corresponding quality rank. The quality rank R**1** represents, for example, quality that is determined to be normal in an initial term **1** during a change in likelihood over time. The quality rank R**2** represents, for example, quality that is determined to be normal in a middle term **2** during the change over time. The quality rank R**3** represents, for example, quality that is determined to be normal in a final term **3** during the change over time.

[0117] The determination unit **102** calculates a plurality of likelihoods based on a plurality of models corresponding to such quality ranks. Typically, when the likelihood calculated based on any one of the models exceeds the threshold, the likelihood calculated based on the other models will not exceed the threshold. When the determination unit **102** calculates a likelihood based on a certain model and the likelihood exceeds the threshold, the determination unit **102** determines the quality rank of this model and outputs the quality rank as a result of determination. When the determination unit **102** calculates the likelihood based on these models and no likelihood exceeds the threshold, the determination unit **102** determines that the operation is performed in an abnormal state.

[0118] The models may be generated in advance for the respective quality ranks. According to the

sixth modification, the determination unit **102** can determine whether the operation is performed in a normal state, and also determine the quality level. The determination unit **102** can also determine the term in the change in likelihood to which the current operation corresponds.

Seventh Modification

[0119] The determination unit **102** may use a plurality of models and change them in accordance with the elapsed time to perform determination. FIGS. **24-1** to **24-3** are diagrams illustrating an example of a determination method according to a seventh modification.

[0120] FIGS. **24-1**, **24-2**, and **24-3** correspond to a term **1**, **2**, and **3**, respectively, and each illustrate an example of the likelihood calculated based on a model generated for the corresponding term. The terms **1**, **2**, and **3** correspond to the initial, middle, and final terms, respectively, in the change in likelihood over time. Each model may be generated in advance based on the sensing information detected during the corresponding term.

[0121] The determination unit **102** determines which model to take from the models for the respective terms based on, for example, context information for specifying the term such as the cumulative operating time. The determination unit **102** determines whether the operation is performed in a normal state based on the specified model. According to the seventh modification, the determination unit **102** can determine whether the current state is a reasonable change over time (normal state) or a state that deviates from the reasonable change over time (abnormal state).

Eighth Modification

[0122] The determination unit **102** may perform determination based on a model specified from a plurality of models that are defined in accordance with the quality ranks. When, for example, a higher quality machining is required for a workpiece, the determination unit **102** uses a model defined for a high quality rank. When, for example, a regular quality machining is required, the determination unit **102** uses a model defined for a lower quality rank than the high quality rank. The determination unit **102** may determine which model to take based on, for example, the context information received by the reception unit **101a** and information received via the acceptance unit **103**.

[0123] The computer program executed by the diagnostic device according to the embodiments above is embedded and provided in a ROM, for example.

[0124] The computer program executed by the diagnostic device according to the embodiments above may be recorded in a computer-readable recording medium such as a compact disc read only memory (CD-ROM), a flexible disk (FD), a compact disc recordable (CD-R), or a digital versatile disc (DVD), as an installable or executable file and may be provided as a computer program product.

[0125] The computer program executed by the diagnostic device according to the embodiments above may be stored in a computer connected to a network such as the Internet and provided by being downloaded via the network. Furthermore, the computer program executed by the diagnostic device according to the embodiments above may be provided or distributed via a network such as the Internet.

[0126] The computer program executed by the diagnostic device according to the embodiments above has a module configuration including the units (such as the communication control unit and the determination unit) described above. As actual hardware, the units are loaded on a main memory when the CPU (processor) reads out and executes the computer program from the ROM, and the units are generated on the main memory.

REFERENCE SIGNS LIST

[0127] **51** CPU [0128] **52** ROM [0129] **53** RAM [0130] **54** Communication I/F [0131] **55** Drive control circuit [0132] **56** Motor [0133] **57** Sensor [0134] **58** Bus [0135] **61** CPU [0136] **62** ROM [0137] **63** RAM [0138] **64** Communication I/F [0139] **65** HDD [0140] **66** Bus [0141] **100** Diagnostic device [0142] **101** Communication control unit [0143] **101a** Reception unit [0144] **101b** Transmission unit [0145] **102** Determination unit [0146] **103** Acceptance unit [0147] **104** Feature

extraction unit [0148] **105** Generation unit [0149] **111** Storage unit [0150] **200** Process machine
[0151] **201** Numerical control unit [0152] **202** Communication control unit [0153] **203** Machine
tool [0154] **211** Sensor [0155] **212** Drive unit

CITATION LIST

Patent Literature

[0156] [PTL 1] [0157] Japanese Laid-open Patent Publication No. 2006-184722

Claims

1. (canceled)
2. A diagnostic system comprising: a diagnostic device; and a process machine, having at least one sensor that measures at least one of acoustic data, acceleration data, or an acoustic emission, as sensing information, wherein the diagnostic device includes a receiver configured to receive sensing information and context information from the at least one sensor of the process machine, the sensing information being information on a physical quantity that varies in accordance with an operation by the process machine, the context information indicating a drive unit of the process machine is driving, the drive unit is included in the process machine; and at least one processor configured to specify an actual machining period based on both of the context information and the sensing information, the actual machining period being a period that a tool of the process machine is machining a workpiece, the tool is driven by the drive unit, determine an abnormal state of the tool using a part of the sensing information, the part being corresponding to the specified actual machining period, and the process machine includes: the at least one sensor configured to detect the physical quantity, and a transmitter configured to transmit the detected physical quantity to the diagnostic device.
3. The diagnostic system according to claim 2, wherein the at least one processor is configured to determine the abnormal state of the tool based on a model stored in a memory of the diagnostic device and corresponding to received context information and using the part of the sensing information; update the model stored in the memory via a machine learning method based on the determined abnormal state; and output a quality rank indicating a quality of a machined workpiece based on a determined state of the tool.
4. The diagnostic system according to claim 2, wherein the at least one processor specifies, based on the context information, the sensing information detected while the tool is performing a certain operation, and determines the state of the tool using the specified sensing information.
5. The diagnostic system according to claim 2, wherein the receiver receives a plurality of pieces of sensing information, and based on, out of the received plurality of pieces of sensing information, the sensing information detected during a term which is specified by the received context information and in which the tool is performing a certain operation, the at least one processor determines the state of the tool in the term.
6. The diagnostic system according to claim 2, wherein the receiver receives a plurality of types of sensing information each corresponding to a different physical quantity, and the at least one processor determines the state of the tool based on, out of the types of the sensing information, the sensing information determined in accordance with the context information.
7. The diagnostic system according to claim 2, wherein the at least one processor is further configured to generate, when context information for which no model has been defined is received, a model corresponding to the context information for which no model has been defined based on the sensing information corresponding to the context information.
8. The diagnostic system according to claim 2, wherein the at least one processor is further configured to receive an input of context information that differs from the context information received by the receiver, and the at least one processor determines the state of the tool based on the received sensing information, and based on a model corresponding to at least one of the context

information received by the receiver and the received context information that differs from the context information received by the receiver.

9. The diagnostic system according to claim 2, wherein the at least one processor determines, based on the received sensing information, a term for use in determining the state of the tool from a term specified by the received context information, and determines the state of the tool using the sensing information detected during the determined term.

10. The diagnostic system according to claim 2, wherein the at least one processor obtains a likelihood of the received sensing information, and determines the state of the tool by comparing at least one of the likelihood and a value indicating a change in the likelihood with a threshold.

11. The diagnostic system according to claim 2, wherein the at least one processor obtains a likelihood of the received sensing information and outputs quality information defined in accordance with the likelihood.

12. The diagnostic system according to claim 11, wherein the at least one processor compares at least one of the likelihood and a value indicating a change in the likelihood with a plurality of thresholds to determine quality information from among a plurality of pieces of quality information, and outputs the determined quality information.

13. The diagnostic system according to claim 2, wherein the context information indicates operating time of the drive unit included in the tool.

14. The diagnostic system according to claim 2, wherein the at least one processor is configured to specify a first time point at which an amplitude of the sensing information changes to exceed thresholds and specify a second time point at which the amplitude changes to fall within the thresholds; extract a period between the first time point and the second time point; and determine the period the tool is machining the workpiece based on information extracted from a period between the first time point and the second time point.

15. The diagnostic system according to claim 2, wherein the at least one processor is configured to determine a state of the tool based on a model corresponding to the received context information, the model being a model of one or more models respectively defined for one or more pieces of the context information.

16. The diagnostic system according to claim 15, wherein the model includes a plurality of models defined in accordance with the context information and quality, and the at least one processor determines the state of the tool with respect to each of the models and outputs information indicating the quality corresponding to the model with which the state of the tool is determined to be normal.

17. The diagnostic system according to claim 15, wherein the model includes a plurality of models defined in accordance with the context information and quality, and the at least one processor determines the state of the tool based on the model corresponding to specified quality and the received context information.

18. The diagnostic system according to claim 2, wherein the context information is received from one of the process machine or an operating unit of the diagnostic device.

19. The diagnostic system according to claim 2, wherein the tool of the process machine is machining the workpiece being the tool of the process machine is in contact with the workpiece.

20. A method of detecting an abnormal state of a tool operating in a process machine, the method comprising: receiving sensing information and context information from at least one sensor of the process machine, the sensing information being information on a physical quantity that varies in accordance with an operation by the process machine, the context information indicating a drive unit of the process machine is driving; specifying an actual machining period based on both of the context information and the sensing information, the actual machining period being a period that a tool of the process machine is machining a workpiece, the tool is driven by the drive unit; and determining an abnormal state of the tool using a part of the sensing information, the part being corresponding to the specified actual machining period.

21. A non-transitory machine readable storage medium storing executable instructions that, when executed by a machine, cause the machine to perform a method of detecting an abnormal state of a tool operating in a process machine, comprising: receiving sensing information and context information from at least one sensor of the process machine, the sensing information being information on a physical quantity that varies in accordance with an operation by the process machine, the context information indicating a drive unit of the process machine is driving; specifying an actual machining period based on both of the context information and the sensing information, the actual machining period being a period that a tool of the process machine is machining a workpiece, the tool is driven by the drive unit; and determining an abnormal state of the tool using a part of the sensing information, the part being corresponding to the specified actual machining period.
